Smart Bug Tracker CLI

CLI-Based Python Project

1. Project Overview

Smart Bug Tracker CLI is a command-line based Python application designed to manage software bug records efficiently. It allows users to add, update, delete, search, filter, and analyze bugs through a structured and user-friendly terminal interface. The system uses Object-Oriented Programming principles and JSON-based data persistence to ensure clean design and reliability.

2. Objectives

The main objectives of this project are to demonstrate Python fundamentals, implement OOP structure, perform CRUD operations, enable data persistence, and create a professional CLI application using an external library.

3. Use Case of the System

This system can be used in small development environments for tracking bugs without complex tools. The primary users of this system can be developers, testers, and students. Typical usage includes reporting bugs, updating status, analyzing critical issues, and tracking workload.

4. Key Features

- Add, View, Update, Delete bug records
- Search bugs by ID or title
- Filter by status and severity
- Sort bugs by priority, severity, and date
- Generate reports (summary, workload, overdue, etc.)
- JSON data persistence
- Duplicate detection
- Overdue bug detection
- Priority-based recommendation system

5. System Design (OOP)

The system is divided into four classes:
Bug (entity), BugManager (logic), StorageManager (data), and CLIApp (interface).
This separation ensures modularity, maintainability, and clarity.

6. Workflow

The program loads data from JSON, displays a menu, accepts user input, processes operations through the manager class, and saves updates back to storage. This cycle continues until the user exits.

7. Project Analysis

Strengths:
• Clean OOP structure
• Persistent data storage
• Professional CLI using external library
• Includes reporting and analytical features

Limitations:
• No graphical interface
• No multi-user support
• Limited duplicate detection
• Uses JSON instead of a database

8. Future Improvements

Future enhancements may include GUI or web conversion, authentication system, database integration, advanced duplicate detection, and exportable reports.

9. Conclusion

The Smart Bug Tracker CLI successfully demonstrates Python programming concepts, OOP design, and data persistence. It fulfills basic requirements and includes practical features that simulate real-world bug tracking systems.